SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

1. Project Information

- **Project Name: Green Giraffe Digital Traceability & Market Access Pilot (Kurima Programme)**
- **Challenge & UNDP Office: UNDP Angola**
- **Document Version: v1.0 – Pilot Impact Measurement Framework**

2. SDG Alignment

- ### SDG 1: No Poverty

Target (1.4): Ensure equal access to economic resources and financial services
Indicator (1.4.2): Proportion of adults with access to financial services

- ### SDG 2: Zero Hunger

Target (2.3): Double agricultural productivity and incomes of small-scale food producers
Indicator (2.3.2): Average income of small-scale food producers

- ### SDG 12: Responsible Consumption and Production

Target (12.3): Halve food loss along production and supply chains
Indicator (12.3.1): Food loss index

- ### SDG 13: Climate Action

Target (13.2): Integrate climate measures into policies and planning
Indicator (13.2.1): Climate-relevant agricultural practices documented and verified

3. Key Performance Indicators (KPIs)

KPI	Measurement Method	Target / Threshold	Notes
Smallholder farmers onboarded	Farmer registry & digital ID logs	1,500 farmers	Disaggregated by gender & province

Average farmer income increase	Sales & escrow settlement records	≥25% increase vs baseline	Verified post-reconciliation
Post-harvest loss reduction	Declared vs reconciled batch quantities	≥30% reduction	Measured per crop cycle
Verified traceable batches	On-chain batch records (Cardano)	100% of pilot batches	Includes QA and lab verification
Payments released through escrow	Multicaixa Express + Stanbic settlement logs	100% of verified batches paid digitally	No cash payments
ESG-complete value chains	ESG & LCA module reports	≥80% of pilot value chains	Covers Environmental, Social, and Governance indicators

4. Tracking Methods

On-chain Metrics (Cardano)

- Batch creation & verification transactions
- QA inspection hashes and timestamps
- Compliance summaries anchored on-chain
- Immutable reconciliation records

Off-chain Metrics

- Agribalyse 3.2 LCA dashboards
- QA & farmer activity logs
- Satellite-based land-use & deforestation checks (NASA HLS, GFW)
- Climate & soil indicators (OpenWeatherMap, iSDAsoil)

Example:
Batch verification data is anchored on-chain at each QA milestone and reconciled monthly with LCA, ESG, and payment dashboards.

5. Baseline vs Projected Outcomes

Metric	Baseline	Projected Outcome
Farmers with formal market access	<20%	≥80%
Average post-harvest loss	~30–40%	≤15%
Digitally traceable batches	0	100%
Verified ESG/LCA reports	0	≥80% of batches
Digital payments to farmers	<10%	100%

6. Monitoring & Reporting Plan

Frequency: Quarterly monitoring; annual impact report

Responsible Party: Green Giraffe M&E Lead + UNDP Angola Programme Team

Format:

- On-chain audit logs
- Interactive dashboards
- PDF impact reports for UNDP & donors

Tools:
Agribalyse 3.2 LCA, Cardano explorers, QA dashboards, satellite data platforms

7. Risks & Mitigation

Risk: Limited digital literacy among farmers

- **Mitigation:** QA-led onboarding, training modules, assisted data capture

Risk: Connectivity gaps in rural areas

- **Mitigation:** Offline-first mobile workflows with delayed sync

Risk: Data inconsistency across actors

- **Mitigation:** QA verification, append-only records, blockchain anchoring

Risk: Delays in payment settlement

- **Mitigation:** Escrow logic with automated reconciliation triggers